

PRINCIPLES OF FIXED ORTHODONTIC APPLIANCES



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LEARNING OUTCOMES

BY THE END OF THIS TOPIC, YOU WILL BE ABLE TO

- **DEFINE FIXED APPLIANCES**
- **DISCUSS THE DEVELOPMENT OF CONTEMPORARY FIXED APPLIANCES**
- **CLASSIFY FIXED APPLIANCE SYSTEM**
- **LIST THE INDICATIONS AND CONTRAINDICATIONS OF FIXED APPLIANCES**
- **IDENTIFY COMPONENTS OF FIXED ORTHODONTIC APPLIANCES**
- **LIST STEPS ON APPLICATION OF FIXED ORTHODONTIC TREATMENT**
- **EXPLAIN THE STAGES OF COMPREHENSIVE FIXED ORTHODONTIC TREATMENT**

INTRODUCTION

- Appliances that are fixed or fitted onto the teeth by the operator and can not be removed by the patient are called **fixed Orthodontic appliances**
- With fixed orthodontic appliances the control over treatment mechanics shifts more directly into the clinicians hands and the patient is restricted to simply maintaining the appliance and oral hygiene
- **Patient compliance** is **rarely** a problem.
- The **control** achieved with fixed orthodontic appliance is far greater as compared to removable appliances and the teeth can be moved in all the **three planes of space**.

ADVANTAGES OF FIXED ORTHODONTIC APPLIANCES

1. *Precise tooth control is possible.*
2. *Multiple tooth movements are possible.*
3. *Patient cooperation is reduced in comparison to removable appliance wear.*

DISADVANTAGES OF FIXED ORTHODONTIC APPLIANCES

1. *Oral hygiene requirement*
2. *Esthetics: Fixed appliances are generally made of metal that might not be esthetically acceptable to the patient.*
3. *Special training for operator: **Only orthodontists are trained to handle and monitor these appliances.***
 - ***Special training is a must to achieve acceptable results.***
4. *Increased cost of treatment*
5. *Increased chair side time*
6. *Anchorage control is more difficult as compared to removable appliances.*
7. *Treatment monitoring is more difficult.*

The Development of Contemporary Fixed Appliances

- Angle's Progression to the Edgewise Appliance
- Edward Angle's position as the "father of modern orthodontics" is based not only on his contributions to classification and diagnosis but also on his creativity in developing new orthodontic appliances.
- With few exceptions, the fixed appliances used in contemporary orthodontics are based on Angle's designs from the early 20th century.



Edward H. Angle in his 50s,

Angle developed four major appliance systems:

1. E-arch. In the late 1800s, a typical orthodontic appliance depended on some sort of rigid framework to which the teeth were tied so that they could be expanded to the arch form dictated by the appliance.

- Bands were placed on molar teeth, and a heavy labial archwire extended around the arch.
- The end of the wire was threaded, and a small nut placed on the threaded portion of the arch allowed the archwire to be advanced so that the arch perimeter increased.
- Individual teeth were simply ligated to this expansion arch.

- Expansion and proclination
- Tipping movements
- Heavy interrupted force

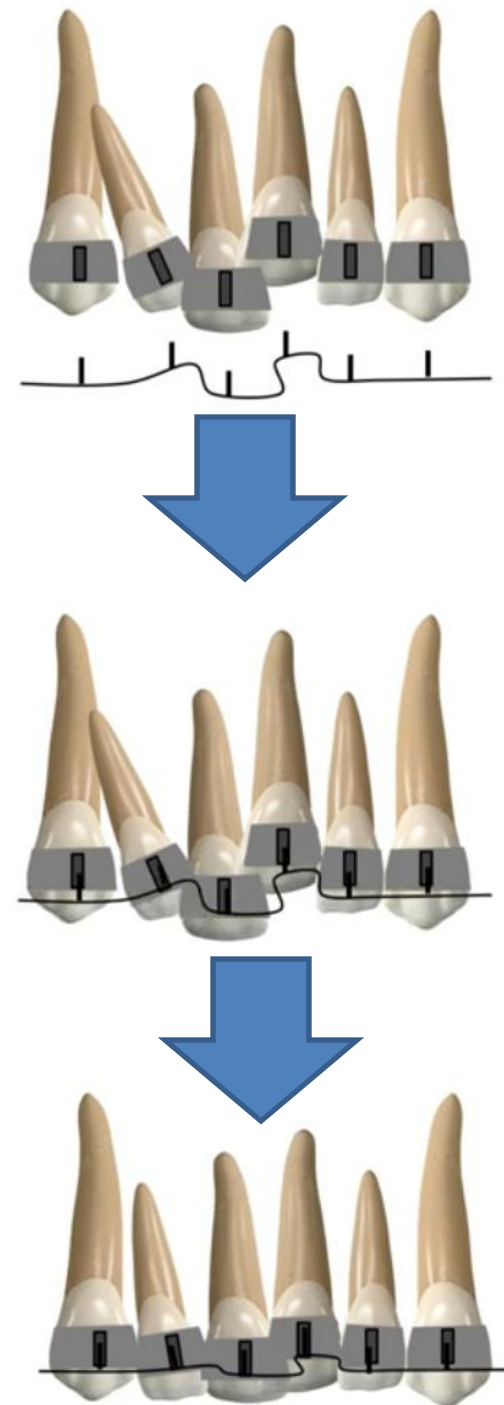


• Molars v by frictic



2. Pin and tube. Angle began placing bands on other teeth and used a vertical tube on each tooth into which a soldered pin from a smaller archwire was placed.

- With this appliance, tooth movement was accomplished by repositioning the individual pins at each appointment.
 - An incredible degree of craftsmanship was involved in constructing and adjusting this pin and tube appliance, and although it was theoretically capable of great precision in tooth movement, it proved impractical in clinical use.
 - It is said that only Angle himself and one of his students ever mastered the appliance.
 - The relatively heavy base arch meant that spring qualities were poor, and the problem therefore was compounded because many small adjustments were needed.
-
- Resoldering of pins were required in each visit.
 - Impractical
 - Heavy interrupted force



3. *Ribbon arch.* Angle's next appliance modified the tube on each tooth to provide a vertically positioned rectangular slot behind the tube.

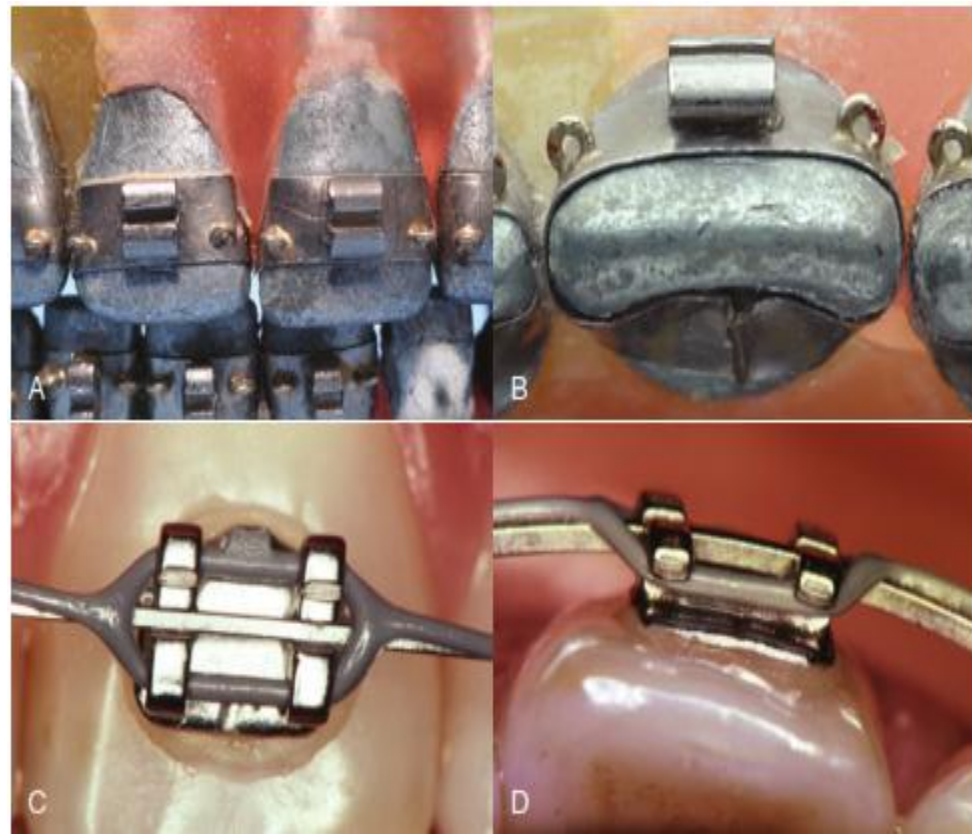
- A ribbon archwire of 10 × 20 gold wire was placed into the vertical slot and held with pins.
- The ribbon arch was an immediate success, primarily because the archwire, unlike any of its predecessors, was small enough to have good spring qualities and therefore was quite efficient in aligning malpositioned teeth.
- Although the ribbon arch could be twisted as it was inserted into its slot, the major weakness of the appliance was that it provided relatively poor control of root position.
- The resiliency of the ribbon archwire simply did not allow generation of the moments necessary to torque roots to a new position

- Gold wire 10x20
- Good Spring properties therefore efficient alignment
- Still root control (moment) needed for torquing and mesiodistal movements was difficult
- Rotations were also difficult



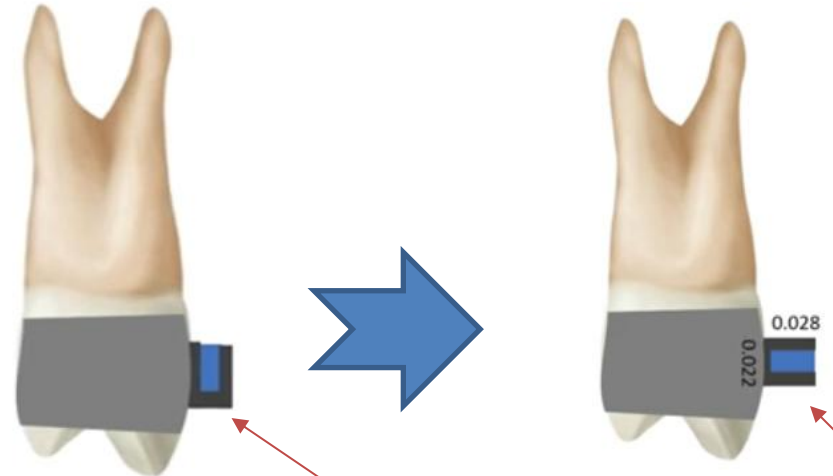
4. Edgewise. To overcome the deficiencies of the ribbon arch, Angle reoriented the slot from vertical to horizontal and inserted a rectangular wire rotated 90 degrees to the orientation it had with the ribbon arch, thus the name “edgewise” .

- The dimensions of the slot were altered to 22 × 28 mil, and a 22 × 28 precious metal wire was used. These dimensions, arrived at after extensive experimentation, did allow excellent control of crown and root position in all three planes of space.



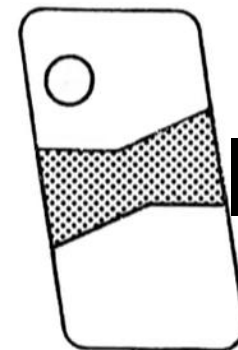
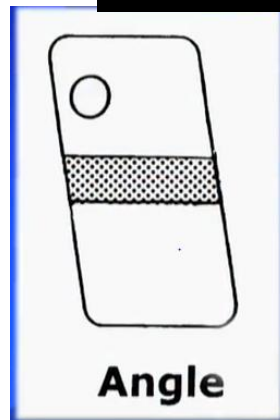
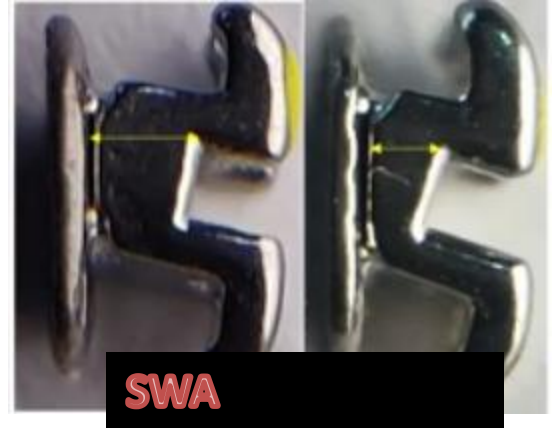
- Dimensions were changed 0.022x 0.028 inches
- Efficient in all 3 dimensional movements
- Some finishing bends were needed

NOT STRAIGHT WIRE APPLIANCE (SWA) OR PRE-ADJUSTED



Fixed orthodontic appliance systems/techniques

- Edgewise
- **PRE-ADJUSTED/SWA**
- **Begg appliance**--round Australian stainless steel wires to allow for the free tipping of teeth, achieving control through the use of auxiliary uprighting and torquing springs
- Tip-edge



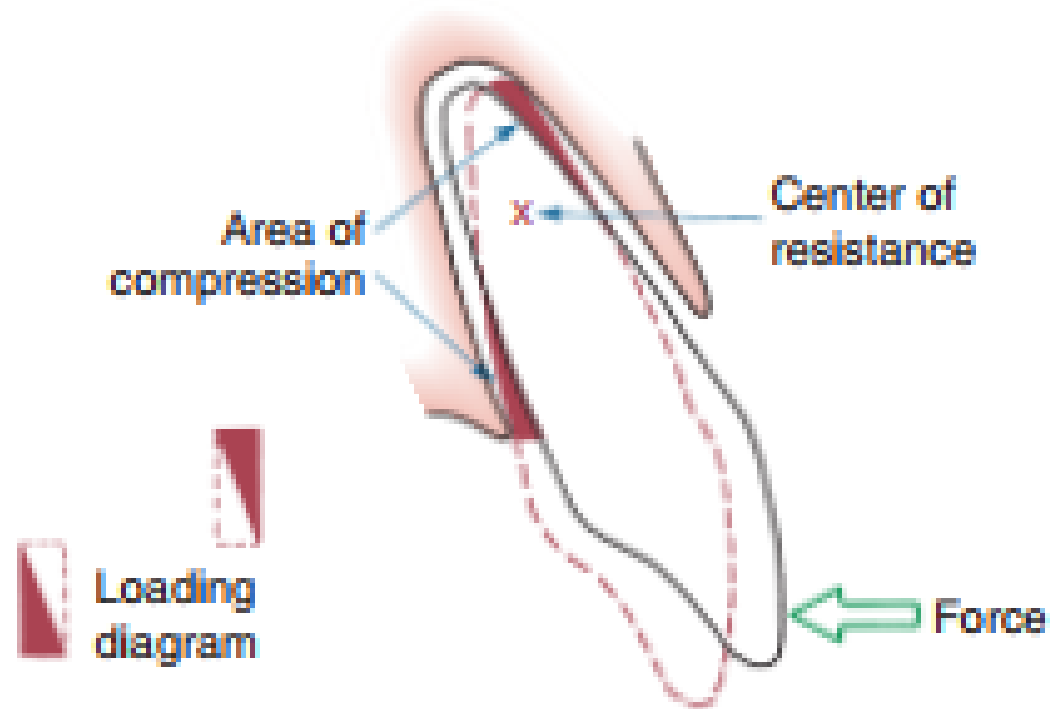
Tip edge



INDICATIONS OF FIXED APPLIANCES

- Fixed orthodontic appliances are indicated whenever **multiple tooth movement** is required, e.g.
 - Rotation
 - Translation
 - Tipping
 - Intrusion
 - Extrusion
 - Torquing
 - Uprighting
 - controlled space closure at extraction sites, etc.

Tipping



Translation

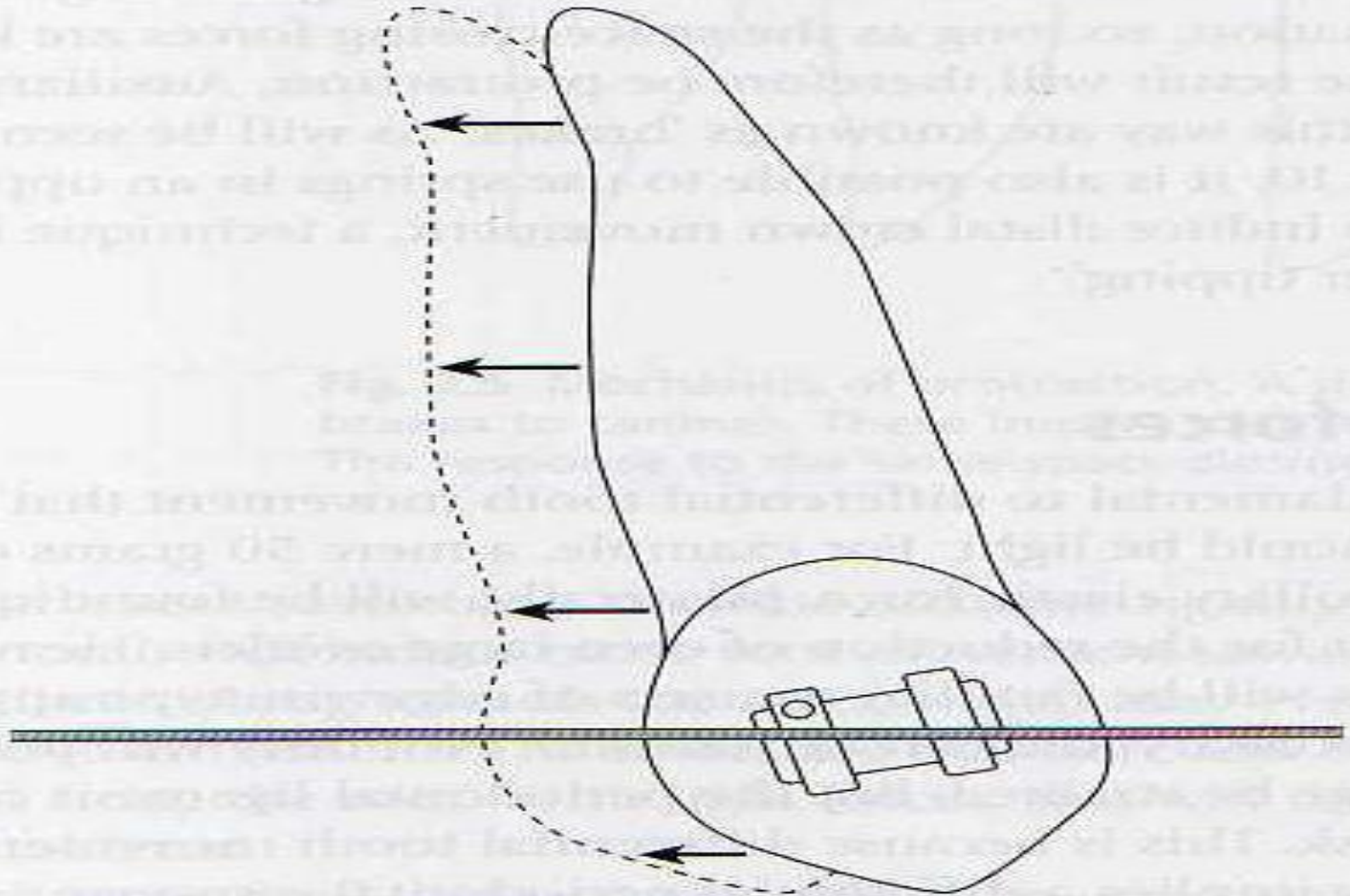
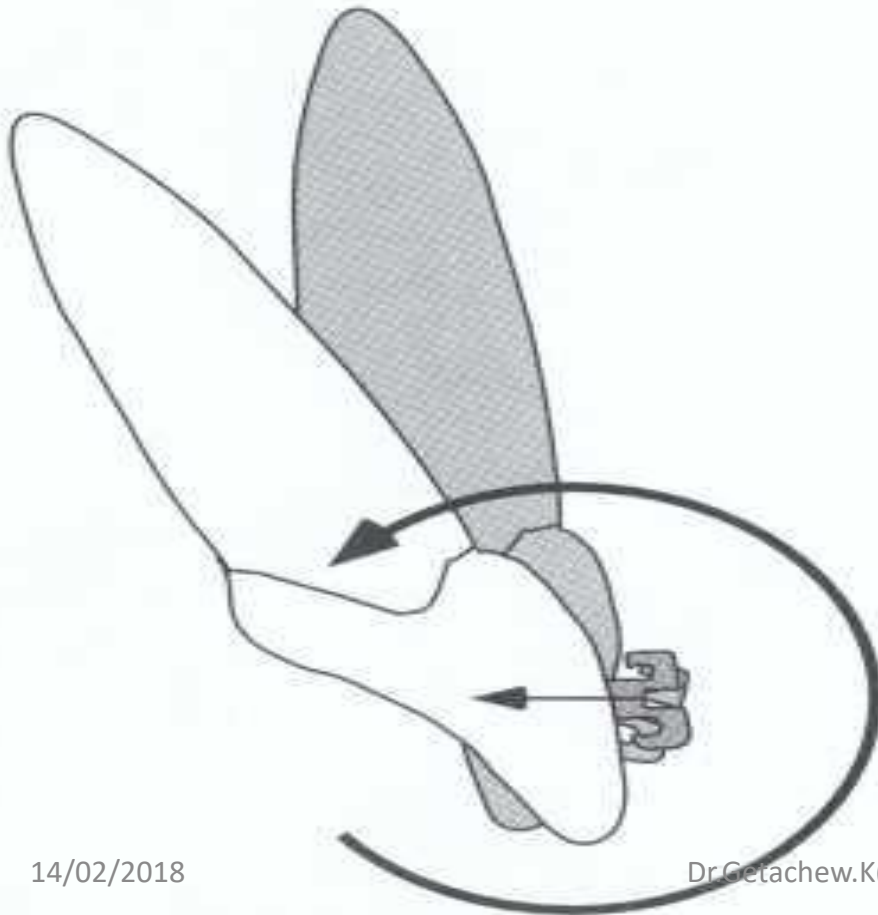


Fig. 2.1 Bodily movement during retraction with an edgewise derived bracket.

Torquing



Uprighting

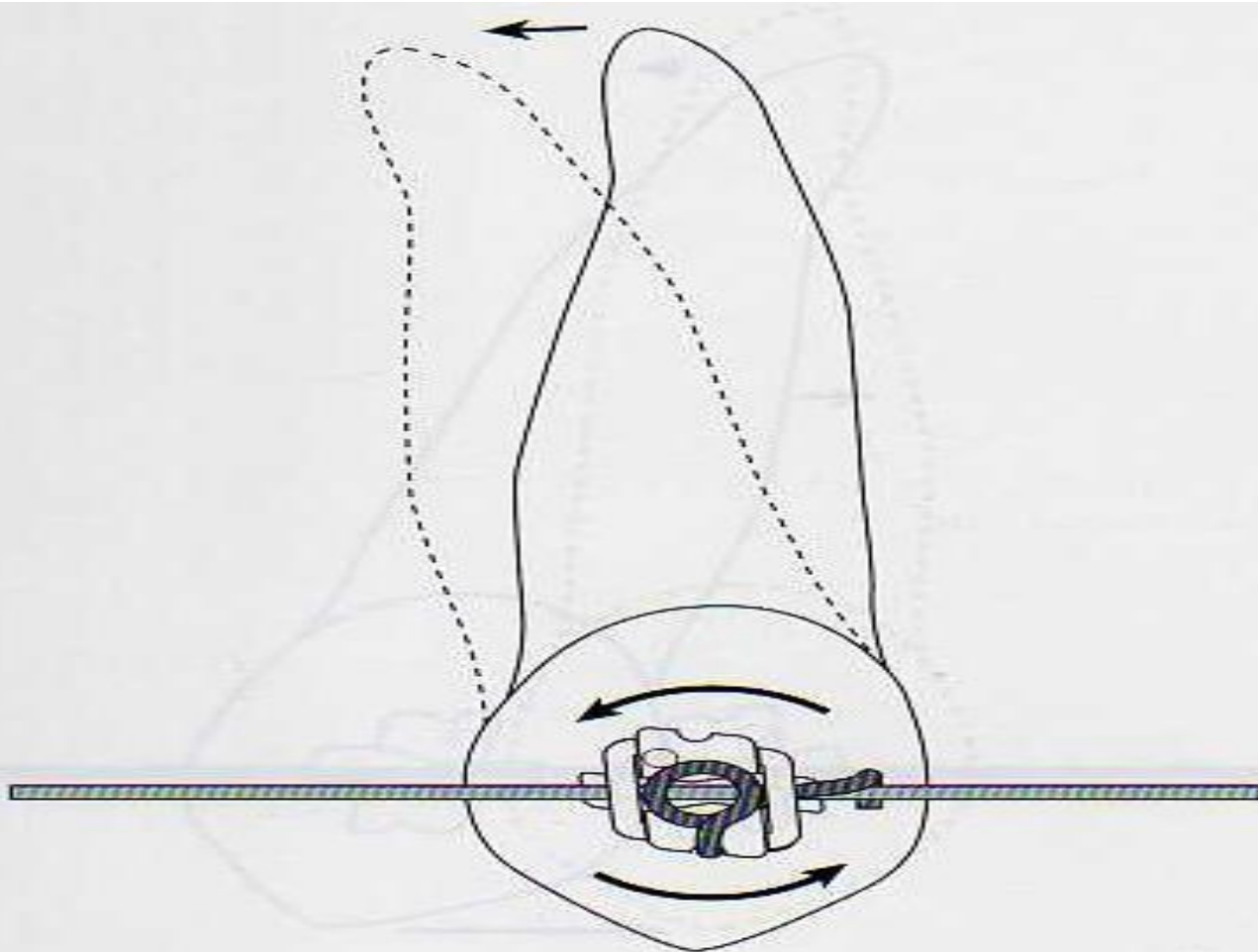
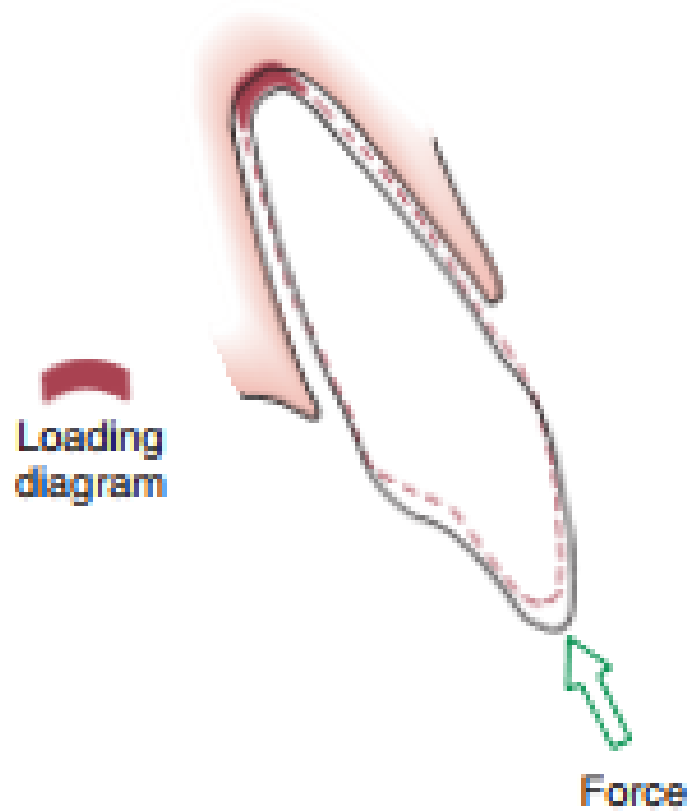


Fig. 2.3 Subsequent root uprighting of a tooth previously tipped during retraction.

Intrusion



Force Levels for tooth Movement

TABLE 8.3 Optimum Forces for Orthodontic Tooth Movement

Type of Movement	Force ^a (gm)
Tipping	35-60
Bodily movement (translation)	70-120
Root uprighting	50-100
Rotation	35-60
Extrusion	35-60
Intrusion	10-20

^aValues depend in part on the size of the tooth: smaller values are appropriate for incisors, higher values for multirrooted posterior teeth.

CONTRAINDICATIONS TO FIXED ORTHODONTIC APPLIANCES

1. *Poorly motivated patient*
2. *Poor dental health*
3. *Malocclusions beyond the scope of fixed orthodontic appliances*
4. *Appropriate training of operator : Only orthodontists are qualified to impart treatment using the fixed orthodontic appliances. Care should be taken by students of dentistry, to only attempt something which is within the preview of their curriculum and should not try behaving as orthodontists without undertaking proper training from a recognized institution.*

COMPONENTS OF FIXED ORTHODONTIC APPLIANCES

- The components which form any fixed orthodontic appliance system can be divided into two categories depending upon their ability to generate forces:

✓ **ACTIVE COMPONENTS**

✓ **PASSIVE COMPONENTS**

ACTIVE COMPONENTS

- These include components which are **capable of generating tooth moving forces**. These include:
 - a. Separators
 - b. Archwires
 - c. Elastics
 - d. Elastomerics
 - e. Springs
 - f. Magnets

A. Separators

- Separators are used to create spaces in between two adjacent teeth, generally for the purpose of banding them.
- Initially *soft brass wires*, wires of 0.5 or 0.6 mm diameter were passed interdentally and twisted to create space.



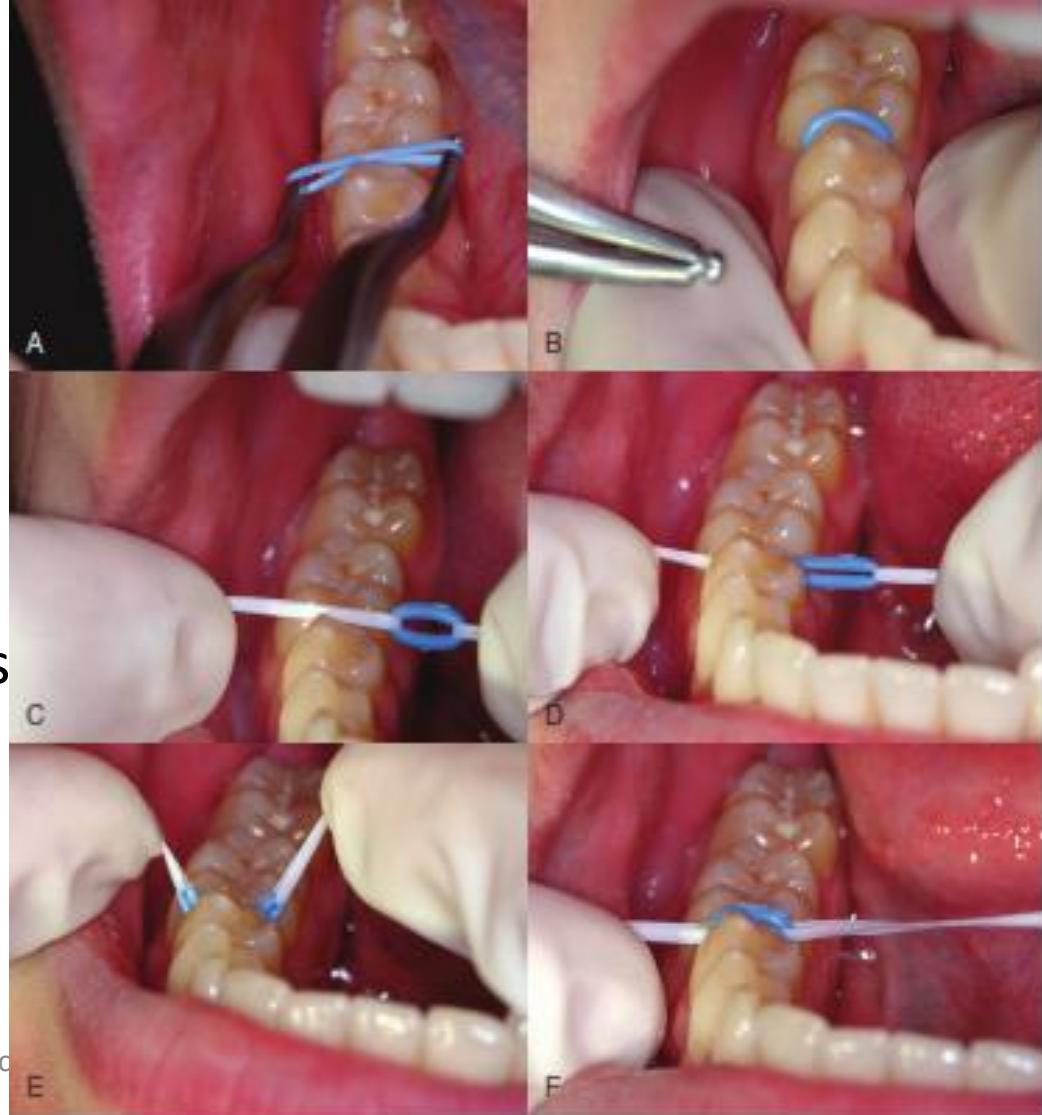
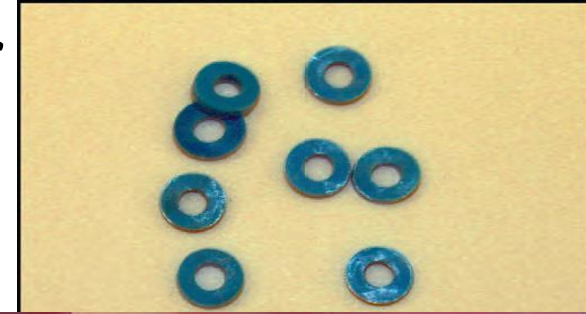
Kesslying separators

- *These are made of 0.016" special plus Australian wire with helices at the closed end*
- The open straight arm passes below the contact sufficient space is created within 48 hours of application.



Ring separators

- *These are small elastic rings* that are stretched and passed through the interdental contacts using the separator placing pliers or using two pieces of dental floss.
- The elastic material used to make these ring separators can be either round or with edges.
- Separation takes about 7 days



Dumb-bell separators

- *These consists of a dumb-bell shaped piece of elastic that is stretched and passed through the contacts between adjacent teeth*
- The elastic of the separator constricts over time to regain its original space and causes separation.



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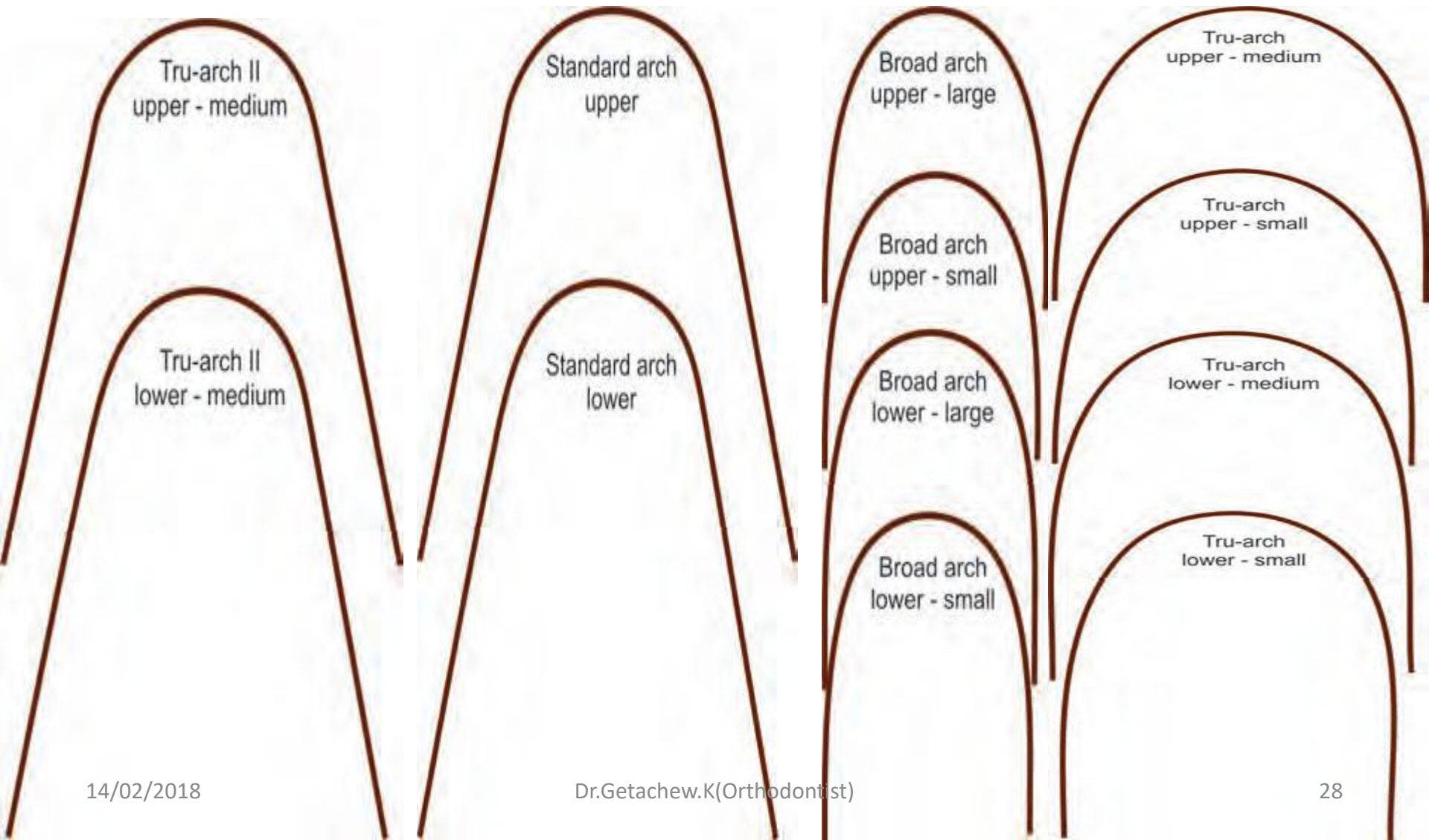


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B. Archwires

- Archwires are available of variable dimensions and materials
- Archwires are available in the prefabricated mould.
- They may be of various shapes depending upon the arch form type.
- The arch form closest to the patient should be chosen unless some change is planned in the arch width.

Various arch forms



C. Elastics

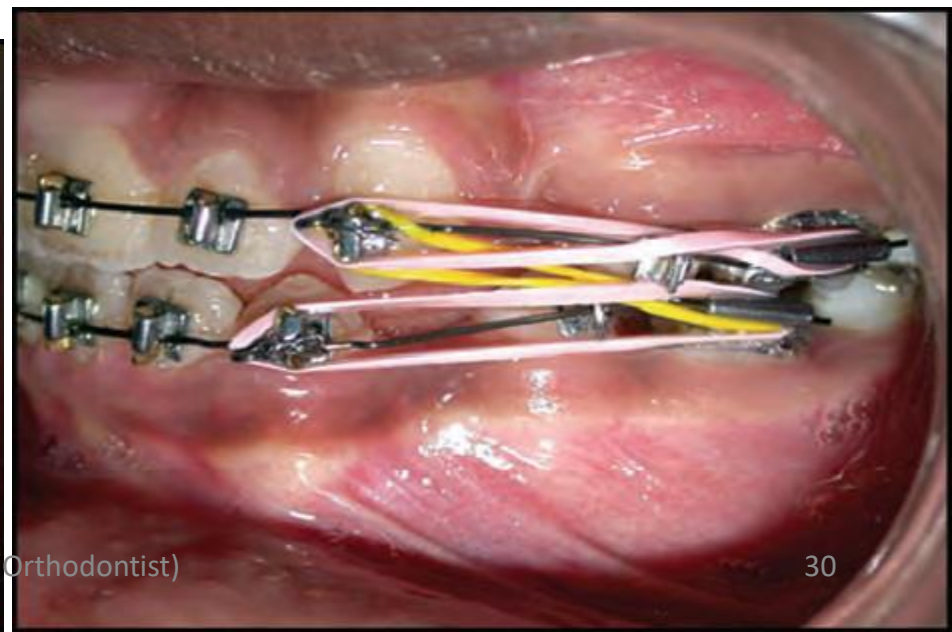
- Elastics can be of latex or non-latex material.
- Non-latex elastics deteriorate less as compared to the latex elastics in the oral environment.
- Elastics are available in various strengths, which is dependent upon their diameter and thickness.
- The elastics are chosen according to the purpose of their use.
- They may be color coded according to strength.



Elastics are used mainly in the following six ways.

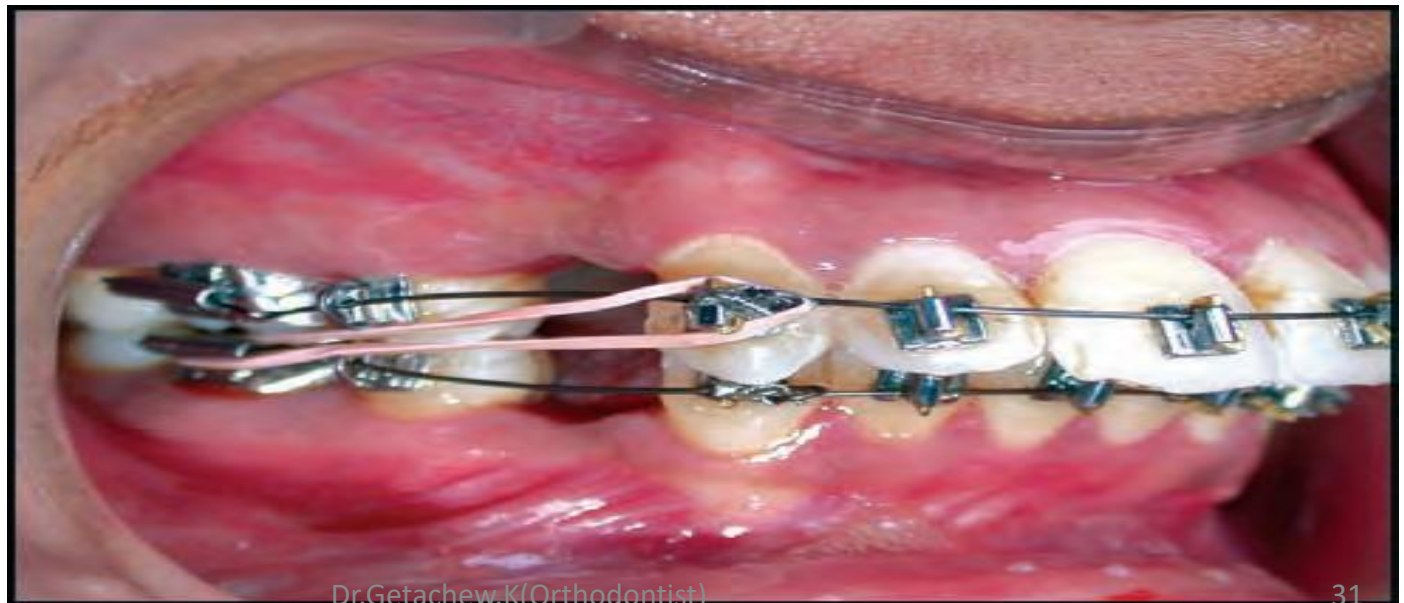
Class I elastics

- These are intra-arch elastics placed mainly between the molars and the anteriors in the same arch.
- They are used to close the extraction spaces by retracting the anteriors (lighter force elastics) or protraction of posterior teeth (heavier force elastics).



Class II elastics

- These are intermaxillary elastics placed between the mandibular molars and maxillary anterior.
- These elastics may be used to produce intrusion of maxillary anterior or decrease the over jet by retracting the maxillary anterior.



Class III elastics

- These are again intermaxillary elastics placed between the maxillary molars and mandibular anteriors.
- These are generally used in the treatment of Angles Class III malocclusions to bring about the retraction of mandibular anteriors and protraction of the maxillary molars.



Diagonal elastics

These are worn generally for the correction of midline deviations.

- The elastic is worn across the anterior teeth diagonally.



Cross bite elastics

- These intermaxillary elastics are used to correct cross bites in the buccal segments.
- Their placement is dependent upon the tooth in cross bite, generally from the palatal surface of the maxillary molars or premolars to the buccal of mandibular molars or premolars.



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Box elastics

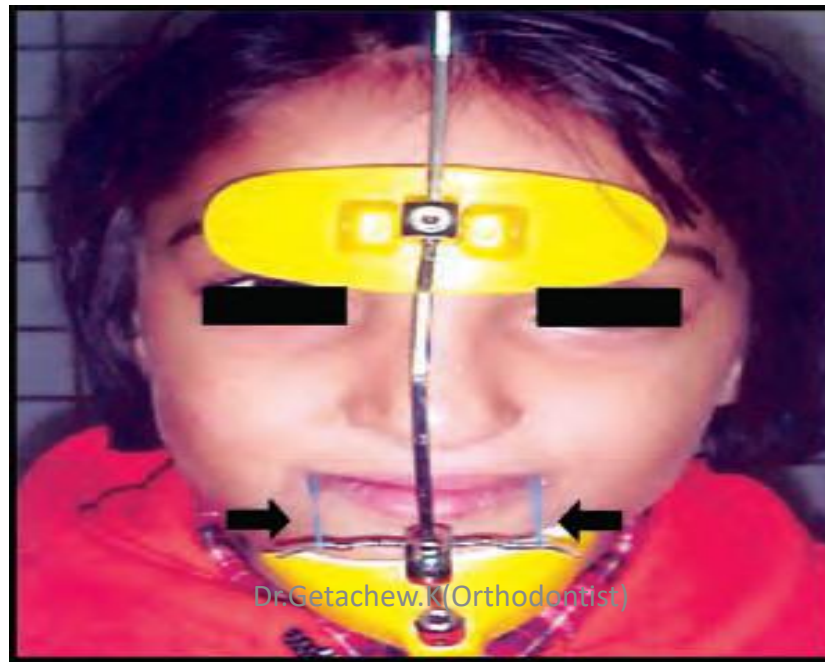
These elastics are used to correct anterior open bites.

- They extend between the maxillary and mandibular anteriors like a 'box', causing the distal tipping of the maxillary anterior and/or forced eruption of the maxillary and/or mandibular anteriors.



Extraoral elastics

- These elastics are used in conjunction with extraoral appliances like a face mask
- Generally exert high forces



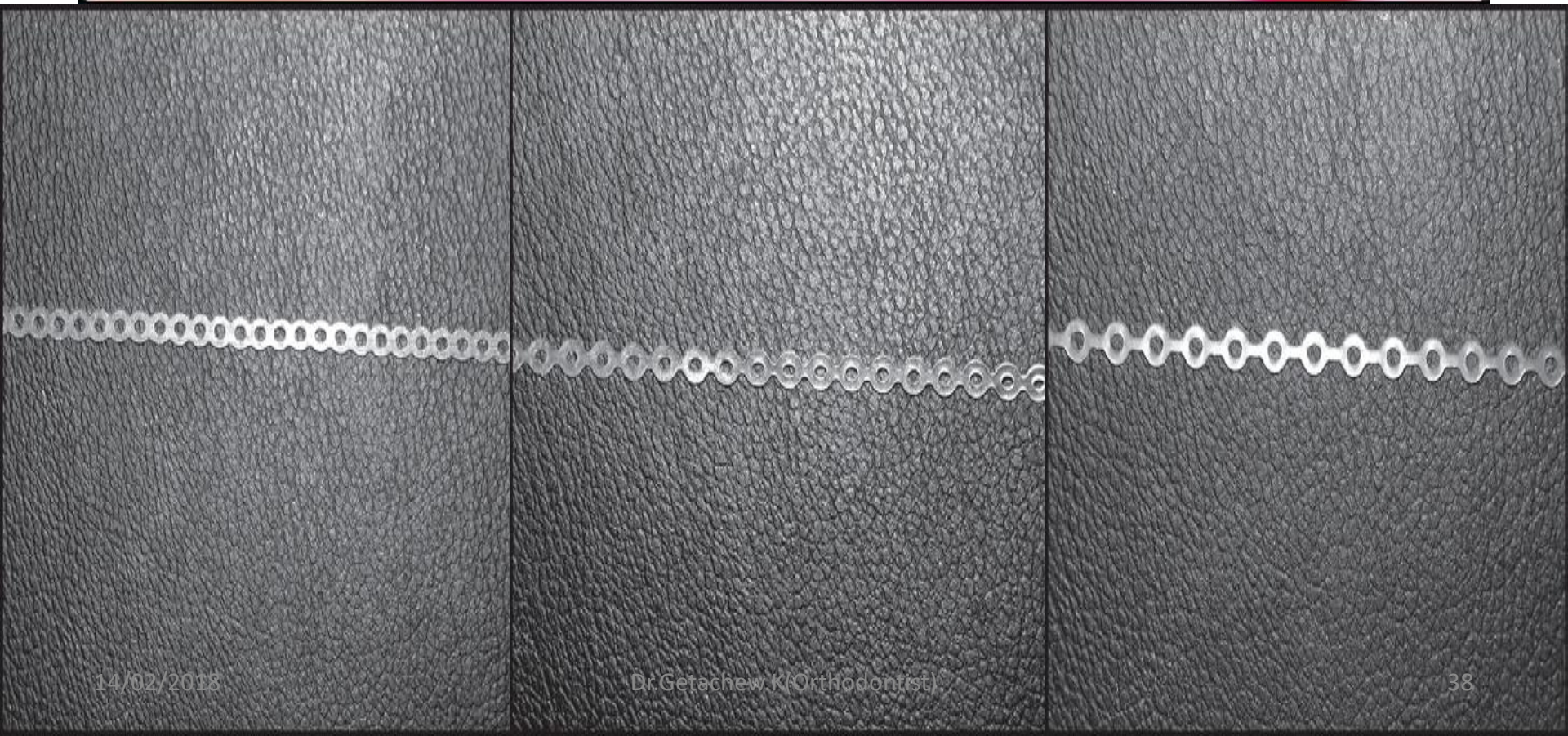
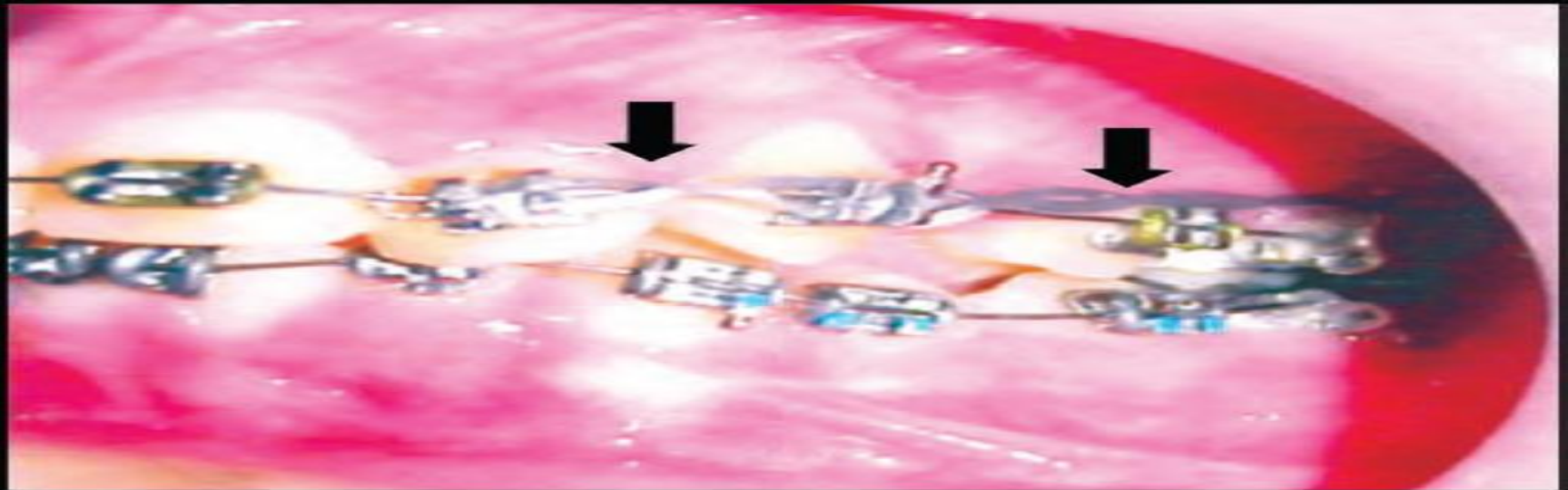
D. Elastomerics

- Elastomerics are used in various forms along with fixed orthodontic appliances.
- These are generally made of synthetic polyurethane materials.

➤ **The various forms of elastomerics used are:**

Elastic chains

- More commonly referred to as E-chains.
- These are mainly used for space closure.
- These are available in three different strengths based on the distance between the rings as continuous or closed short and long



Elastic thread or cotton thread

Made of special elasticized cotton, it is used to exert forces, which are used to correct derotations, consolidation of anterior spacing, etc.



Elastic ligatures

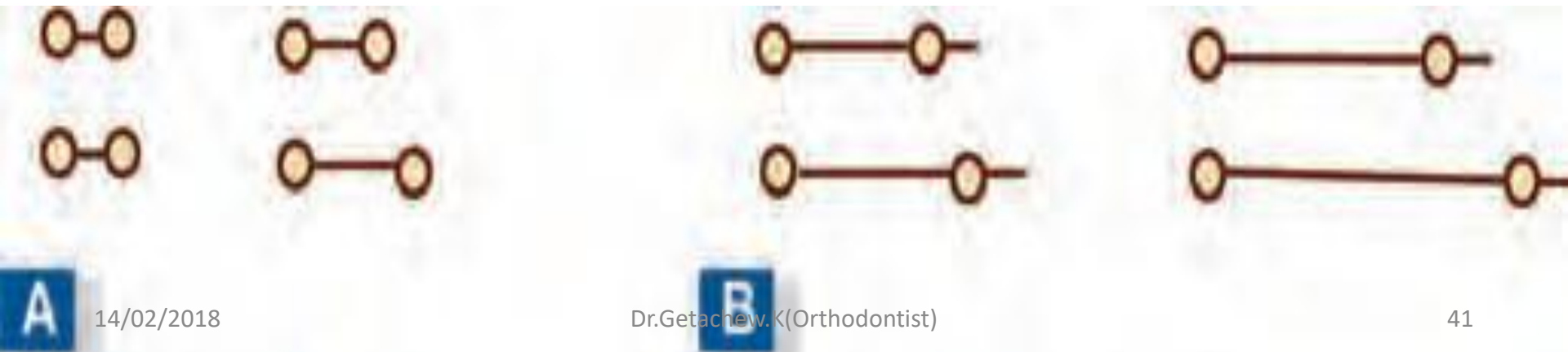
These are used to secure the archwire in edge wise or pre-adjusted edgewise brackets.

- These can be easily engaged under bracket tie wings.
- They are available in various attractive colors



Elastic modules or elastomeric links

- They are made of two elastic rings separated by a variable distance.
- They are available in variable sizes based on the interring distance.
- These are generally used to closed spaces or for derotation of teeth.



E. Springs

- Various types of springs are used as auxiliaries to generate tooth moving forces.
- Springs are mainly used for tooth uprighting and torquing
- Springs may be used to open spaces (open coil springs) or to close spaces (closed coil springs).

Springs used along with the fixed orthodontic appliances are:

- Uprighting springs
- Rotating springs
- Torquing springs
- Open coil springs
- Closed coil spring

Uprighting springs

- Generally made of 0.012" or 0.014" Australian wire.
- They move the tooth root in a mesial or distal direction.



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Side-winder

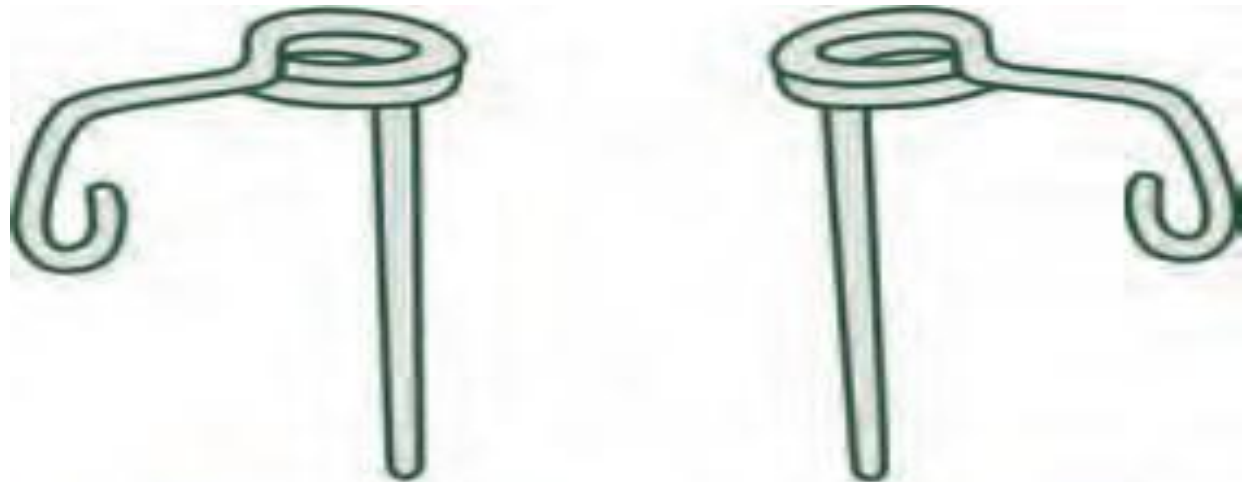
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Standard

Rotating springs

- Rotating springs, as the name suggests, provide for a simple and effective means of derotating teeth without the removal of the archwire.

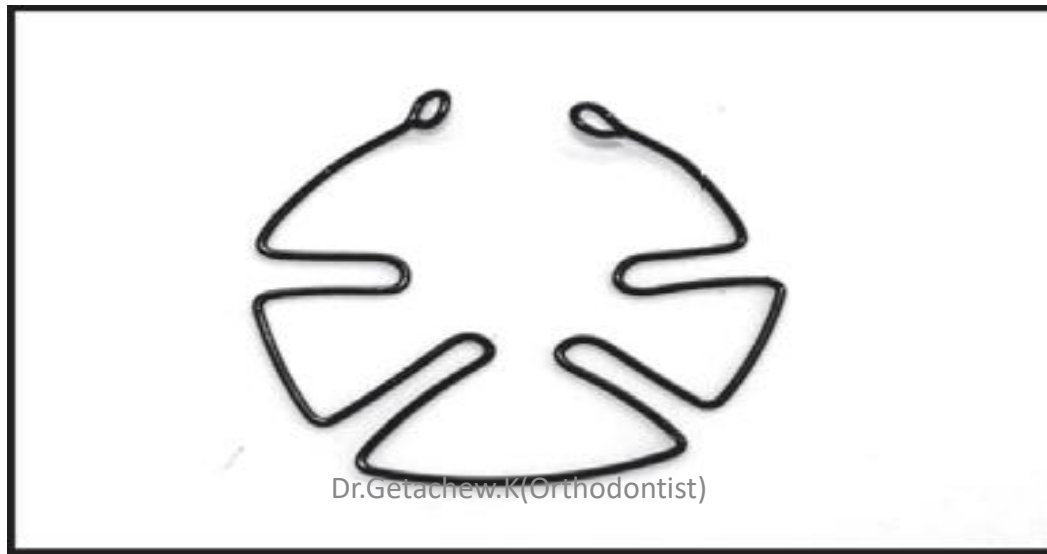


**Clock
wise**

**Counter-
clockwise**

Torquing springs

- Torquing springs are usually made of 0.012" or 0.014" Australian wire.
- They are capable of moving the tooth roots in a labial or lingual/palatal



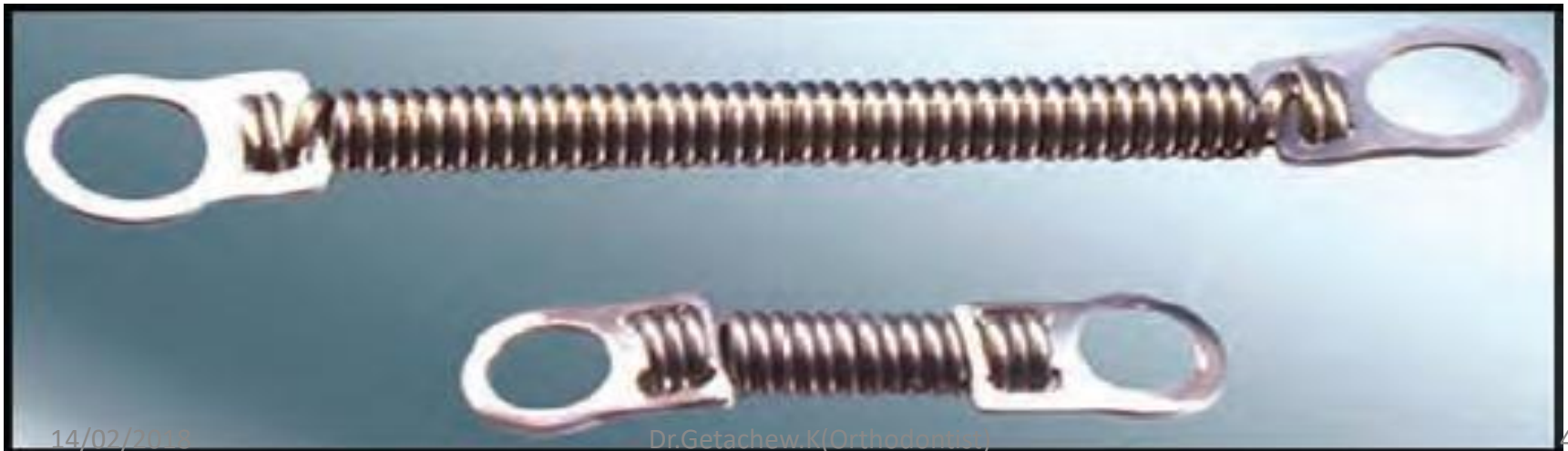
Open coil springs

- Open coil springs are either made of stainless steel or nickel titanium alloy.
- The force exerted also, depends upon the diameter of the wire of which the spring is made.



Closed coil springs

- Closed coil springs can be made of stainless steel or nickel titanium alloys.
- They are used to close spaces.



F. Magnets

- Magnets have been used along with the fixed orthodontic appliances for the purpose of space closure as well as regaining lost space.
 - For the purpose of space closure they are used in attraction mode and for regaining lost space in repulsion mode.
- **Magnets used presently are:**
- Samarium cobalt magnets— SmCo_5 and $\text{Sm}_2\text{Co}_{17}$
 - Neodymium iron boron magnets— $\text{Nd}_2\text{Fe}_{14}\text{B}$

PASSIVE COMPONENTS

- These are those components of the fixed appliances which are not capable of generating tooth moving forces but help in providing attachment for other auxiliaries to the tooth or retaining other active components of the appliances.

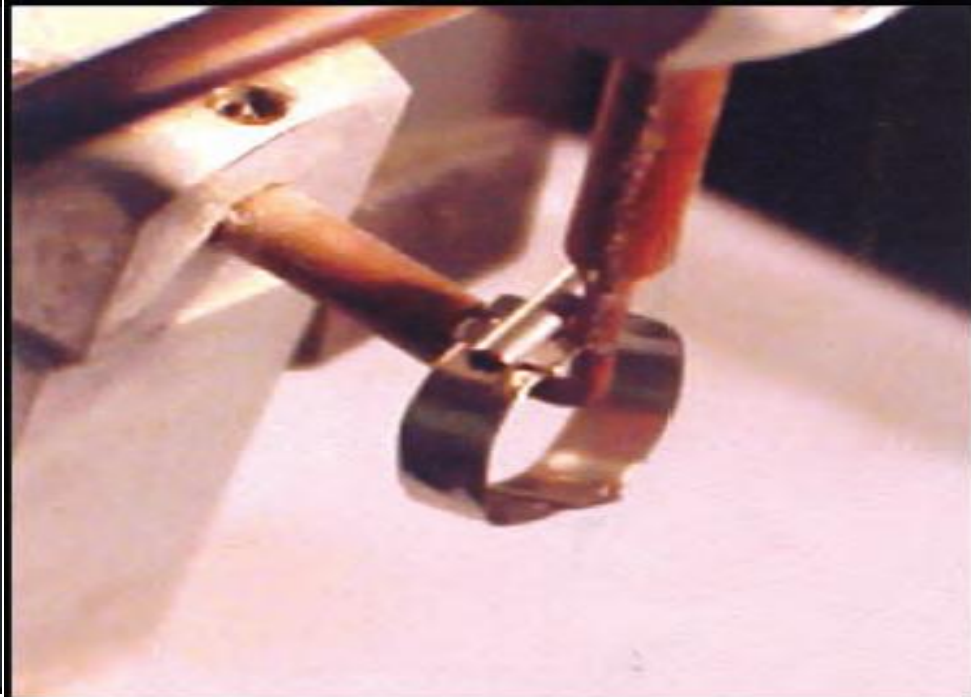
➤ **These include:**

- A. Bands
- B. Brackets
- C. Buccal tubes
- D. Lingual attachments
- E. Lock pins
- F. Ligature wires

A. Bands

Indications of
banding

- These are metal attachments that are cemented to individual teeth and provide a place for attachment of other auxiliaries like buccal tubes, lingual buttons, etc.
- Banding is preferred on teeth that are likely to experience excessive forces, e.g. mandibular molar, mandibular premolars, maxillary second molars or anterior teeth in cross-bite.
- Banding might also be employed in teeth that have large metal restorations (structurally weak) or having metal prosthesis/ crowns (bondings is difficult or weak).



B. Brackets

- A bracket is defined as a device that projects horizontally to support auxiliaries and is open on one side usually in the vertical or horizontal.
- Brackets can either be welded to bands, which are then cemented to individual teeth or can be bonded.

Ss
Titanium
Ceramic
Plastic
Self ligating

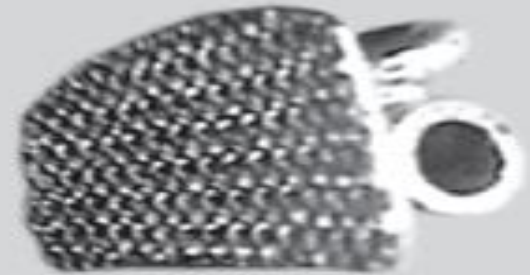
C. Buccal Tubes

- Buccal tubes are horizontal hollow tubes, round, rectangular or oval in shape.
- They are generally used on molars and help provide better three dimensional control of these anchor teeth.

➤ **They can be classified as:**

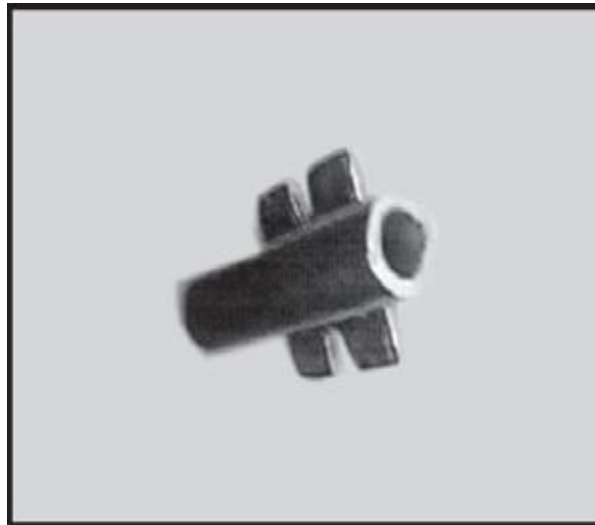
a. Classification based on mode of attachment

- Weldable—can be welded on-to bands
- Bondable—bonded directly to the tooth surface



b. Classified according to lumen shape as

- Round
- Oval
- Rectangular



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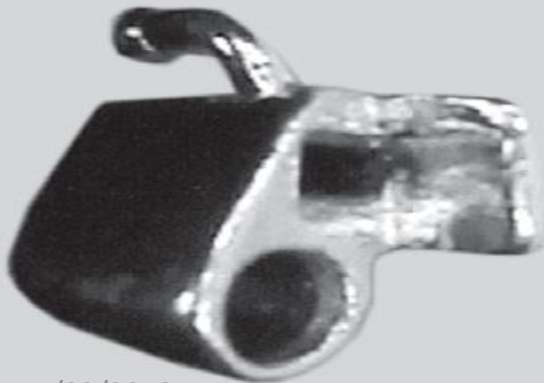
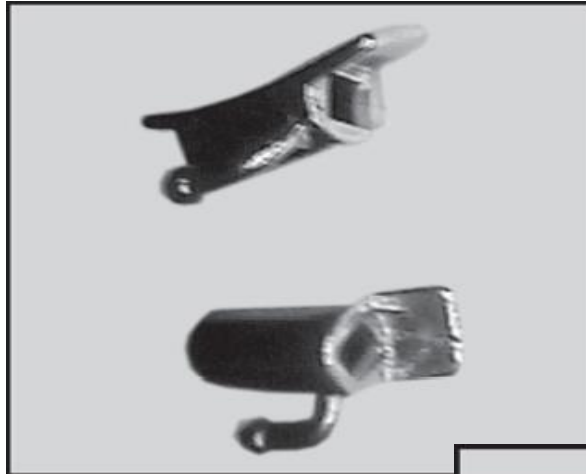
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c. According to the number of tubes

- Single
- Double
- Triple



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D. Lingual Attachments

- These are accessory attachments other than brackets and tubes which are placed on the lingual aspect of teeth (bonded) or bands (welded).

➤ **These include:**

- Lingual buttons are Buttons with variable shaped bases for attachment of elastics or elastomerics.*
- Lingual sheaths These are used for attaching accessories such as transpalatal arches, Ni-Ti molar rotators and expanders.*



E. Lock Pins

- Lock pins are either made of brass or soft steel.
- These are used to hold the wire in bracket slots in the Begg technique or accessories in the tip edge technique.



One-point
safety pin



High hat
safety pin



Double
safety pin



Hook
pin



Universal
T-pin

F. Ligature Wires

- Ligature wires are soft stainless steel wires of 0.008 to 0.010 inch in diameter.
- These may be used to hold / ligate the archwire in brackets or to tie segments of teeth together



STEPS ON APPLICATION OF FIXED ORTHODONTIC TREATMENT

1. Orthodontic banding

- ☐ **first molar** is the most common used teeth for banding .
- ☐ Commonly used separators are made of an **elastomeric material**
- ☐ The most commonly used cementing material in contemporary practice is **glass ionomer cement**

Steps of banding

Step I Separation of adjacent teeth can be achieved using separators for a duration of 2 to 7 days depending upon the separator used.

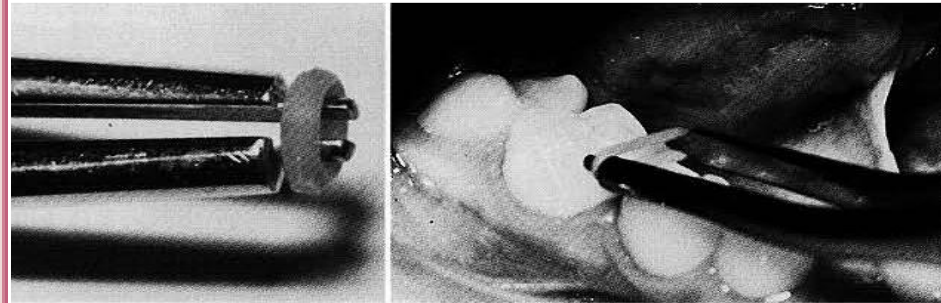
Step II Selection of band material is critical. Thinner band material lesser in width is used for anterior (0.010×3.80 mm) teeth and thicker band material broader in width is used for premolars (0.12×4.55 mm) and molars (0.15×4.55 mm).

Step III Pinching the band—Band material of adequate length is cut and welded at the ends (Fig. 39.25A). The band is adapted around the teeth using a band pusher (Fig. 39.25B) and pinched using the band pinching pliers or a Howe plier (Fig. 39.25C). The pinched band is removed from the tooth and welded close to the pinched ends. Extra band material is cut and bent in a distal direction and welded (Fig. 39.25D). The margins are smoothened and gingival contouring done on the mesial and distal gingival margins using a dental stone (Fig. 39.25E).

Step IV Fixing the attachments—Attachments can be welded or soldered in place. Welding is more commonly performed (Fig. 39.25F)

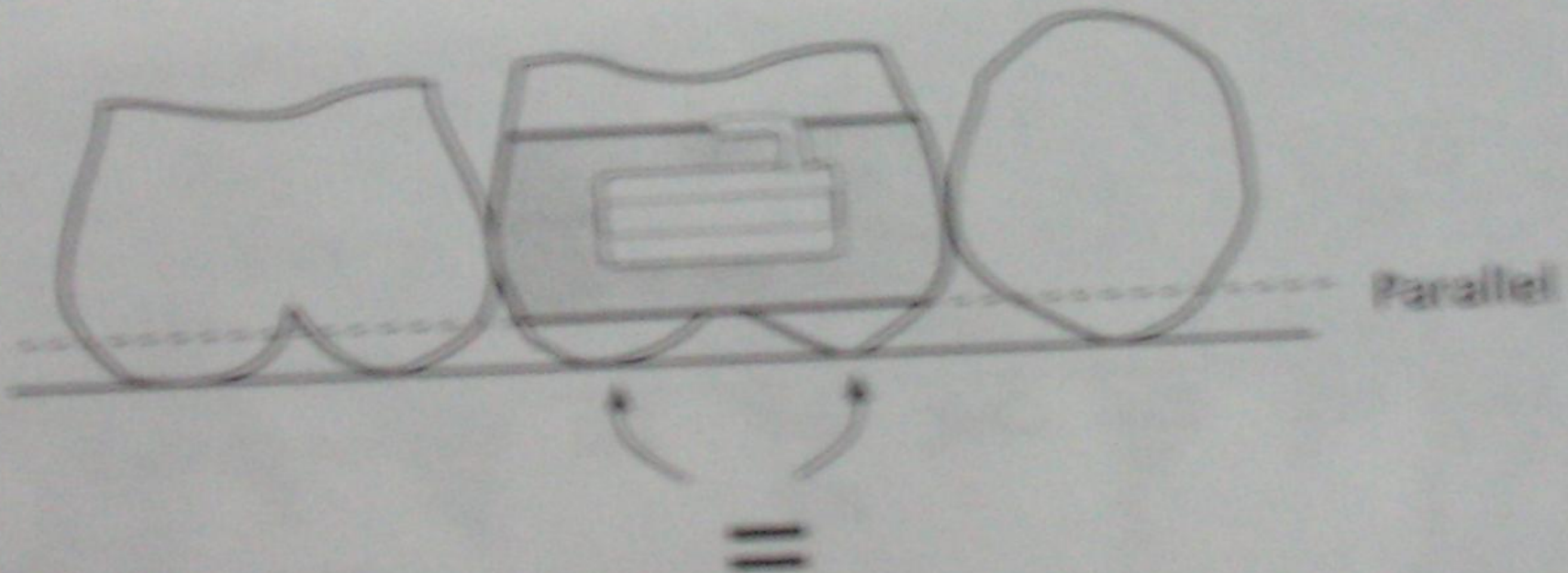
Step V Cementation of the band can be done using any of the luting cements. Adequate moisture control is essential during cementation. Excess cement can be removed using an explorer after the initial set of the luting cement.

Separators



UPPER MOLAR TUBE PLACEMENT:

- The upper molar tube should straddle the buccal groove and care is needed to prevent the distal aspect of the band from seating to gingivally.
- The band should be checked from buccal to ensure it is parallel with buccal cusps.
- It is helpful if the tube is welded more to occlusal, especially for the second molar.



2. Orthodontic bonding

❖ ***Bonding*** is defined as the attachment of the bracket using bonding resins to the enamel surfaces.

□ There are certain basic steps that bonding of orthodontic brackets involves:

Table 39.2: Steps in bonding

Step I Cleaning is done using water slurry of pumice with a rubber cup or polishing brush (Fig. 39.32A)

Step II Moisture control is achieved using cheek and tongue retractors in conjunction with cotton rolls and saliva ejectors

Step III Enamel conditioning is generally achieved using 27 percent phosphoric acid in gel or liquid form for 15 to 30 sec. (Fig. 39.32B). After which the etched surface is washed and dried leaving behind a white frosty appearance (Fig. 39.32C)

Step IV Sealing agent is painted over the entire etched enamel surface using a small form pellet on brush with a single gingivo-incisal stroke (Fig. 39.32D)

Step V A coat of sealant is also applied on the bracket base. This is followed by a sufficient amount of composite being spread on the bracket base. The bracket is accurately placed under pressure into the tooth surface (Fig. 39.32E). Excess flash is removed using a sickle scaler (Fig. 39.32F). Brackets should appear distinct after bonding (Fig. 39.32G)

DIRECT BONDING OF BRACKETS:

After cleaning of the enamel surface etching and application of primer the positioning of brackets are carried out in five steps-

1. The brackets are positioned at the estimated mid-point of the clinical crown with the wings parallel to the long axis of crown.



The bracket is then pressed $\frac{1}{3}$ of the way on to the tooth surface at this position.

2. Excess bonding agent is then removed.



3. Vertical positioning is checked with a gauge.

4. Rotational and horizontal positioning is

re-checked and then bracket is pressed fully.

5. Any additional excess bonding material is removed before light curing.



Relative placement of the brackets on the maxillary and mandibular teeth

❖ MAXILLARY TEETH BRACKET POSITIONING FROM THE INCISAL EDGE

- Central incisors 4.0 mm
- Lateral Incisors 3.5 mm
- Canines 4.5 mm
- Premolars and molars 4.0 mm

❖ MANDIBULAR TEETH BRACKET POSITIONING FROM THE INCISAL EDGE

- Central incisors 3.5 mm
- Lateral Incisors 3.5 mm
- Canines 4.5 mm
- Premolars and molars 4.0 mm

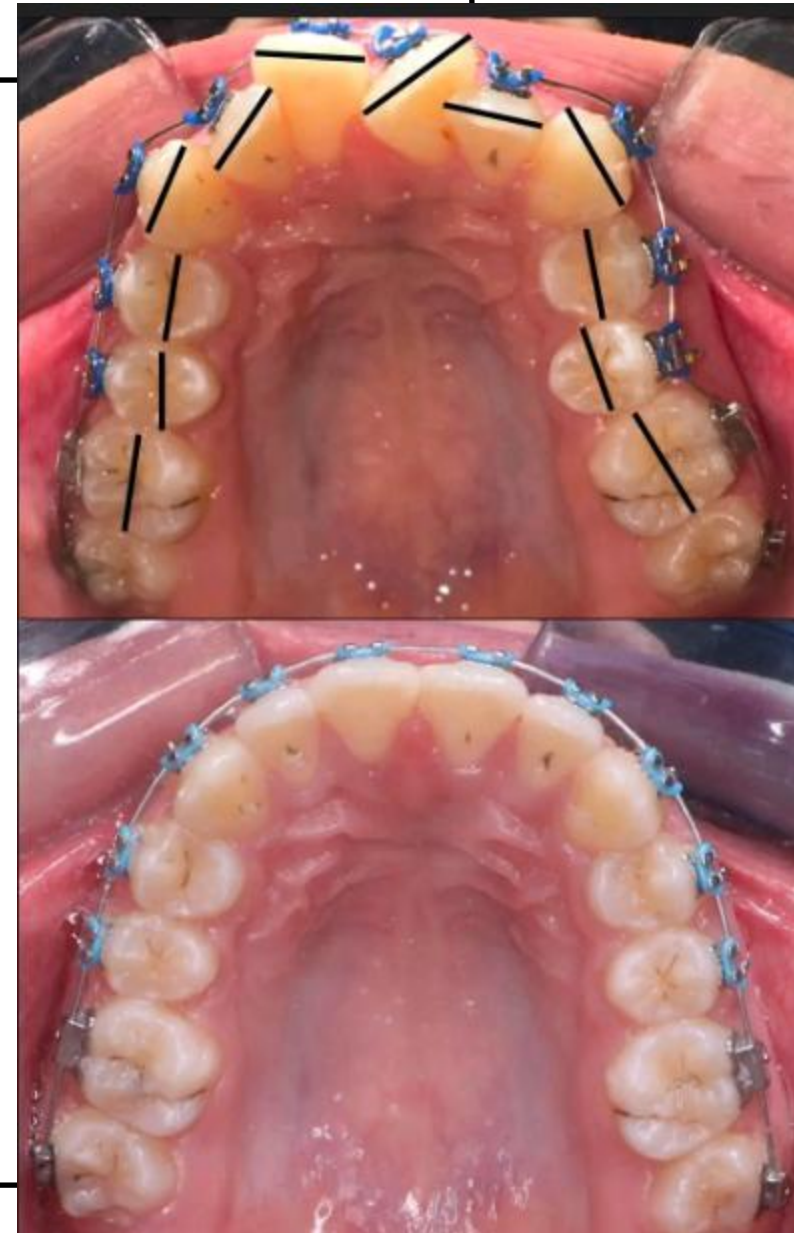
STAGES OF COMPREHENSIVE FIXED ORTHODONTIC TREATMENT

- ❑ There are three comprehensive stages in fixed orthodontic treatment
- 1. The first stage: Alignment & Leveling
- 2. The second Stage: Correction of Molar relationship and Space Closure.
- 3. The Third Stage: Finishing

The first stage: Alignment & Leveling

A. Alignment

- ❑ Is a process that organizes all the teeth into a smooth curve.
- ❑ The major problem is that there is not enough room in the mouth for all of the teeth to align.
- ❑ To bring all the teeth in to alignment a combination of labiolingual and mesiodistal tipping guided by an arch wire is needed but root movement usually is not indicated.
- ❑ The process of alignment is accomplished by uncontrolled tipping of teeth into a smooth arch.



Important consideration During Alignment

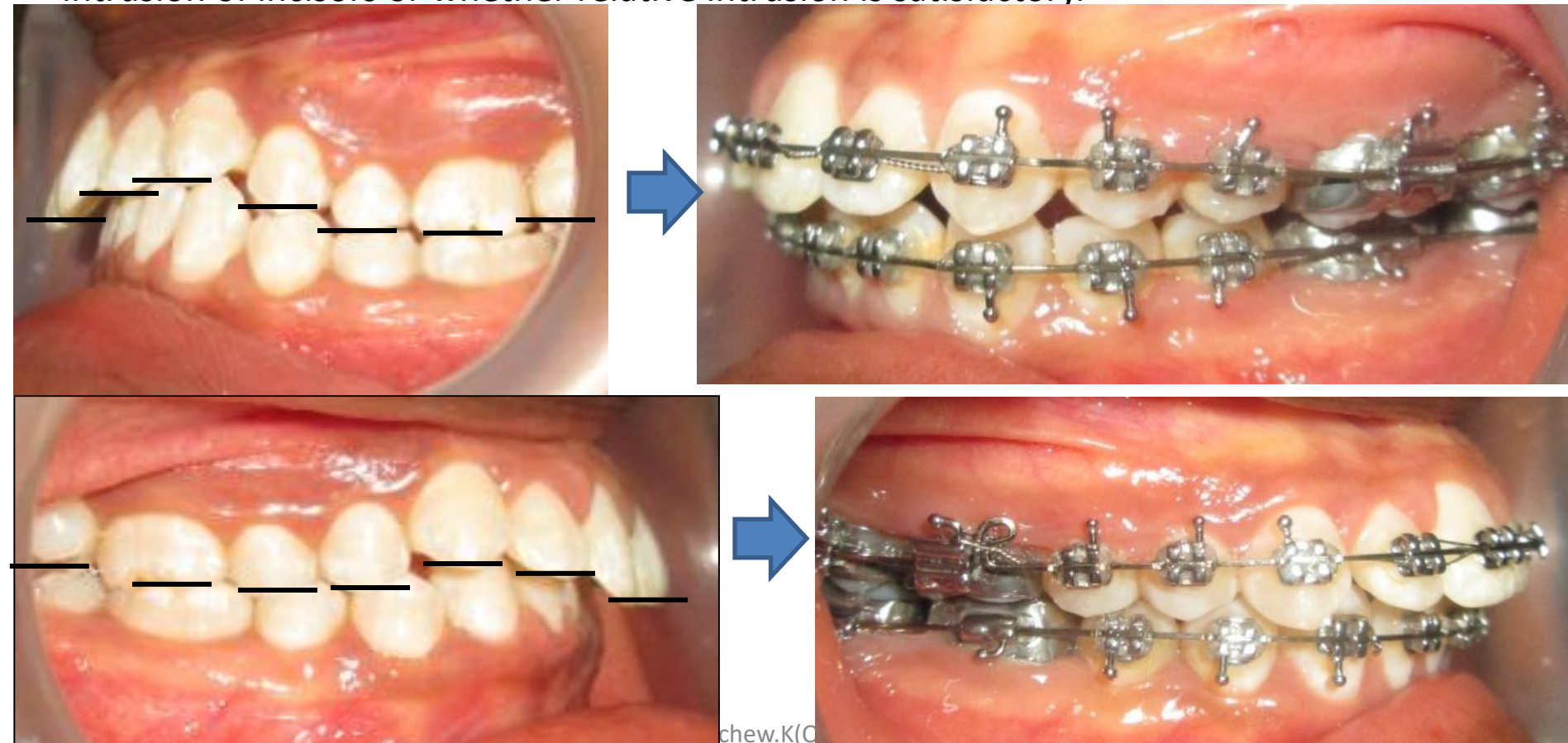
1. **Heavy force should be avoided.**
2. **The arch wires should be able to move freely within the bracket slot.**
3. **Rectangular wire for alignment should be avoided.**
4. **The springiest arch wire are more important for alignment :**

Properties of alignment arch wires

- ❖ The wires for initial alignment requires a combination of
 - Excellent strength (high strength)
 - Excellent springiness (low stiffness)
 - Long range of action (high range)
- ❖ The titanium based arch wires both Nickel-Titanium (NiTi) and Beta Titanium offer a better combination of strength and springiness than do steel wires.
- ❖ The NiTi wires are both **springier** and **stronger** in small cross section than **Beta -Ti**.
- ❖ NiTi wires are particularly useful in the first stage of treatment.

B. LEVELING

- **Leveling** is aligning the upper teeth properly with the lower teeth.
- The challenge of leveling is to make the upper and lower teeth fit together.
- By making the biting surfaces all at the same level, it is possible to make the upper and lower teeth fit together.
- The arch wire designed for leveling depends on whether there is a need for absolute intrusion of incisors or whether relative intrusion is satisfactory.



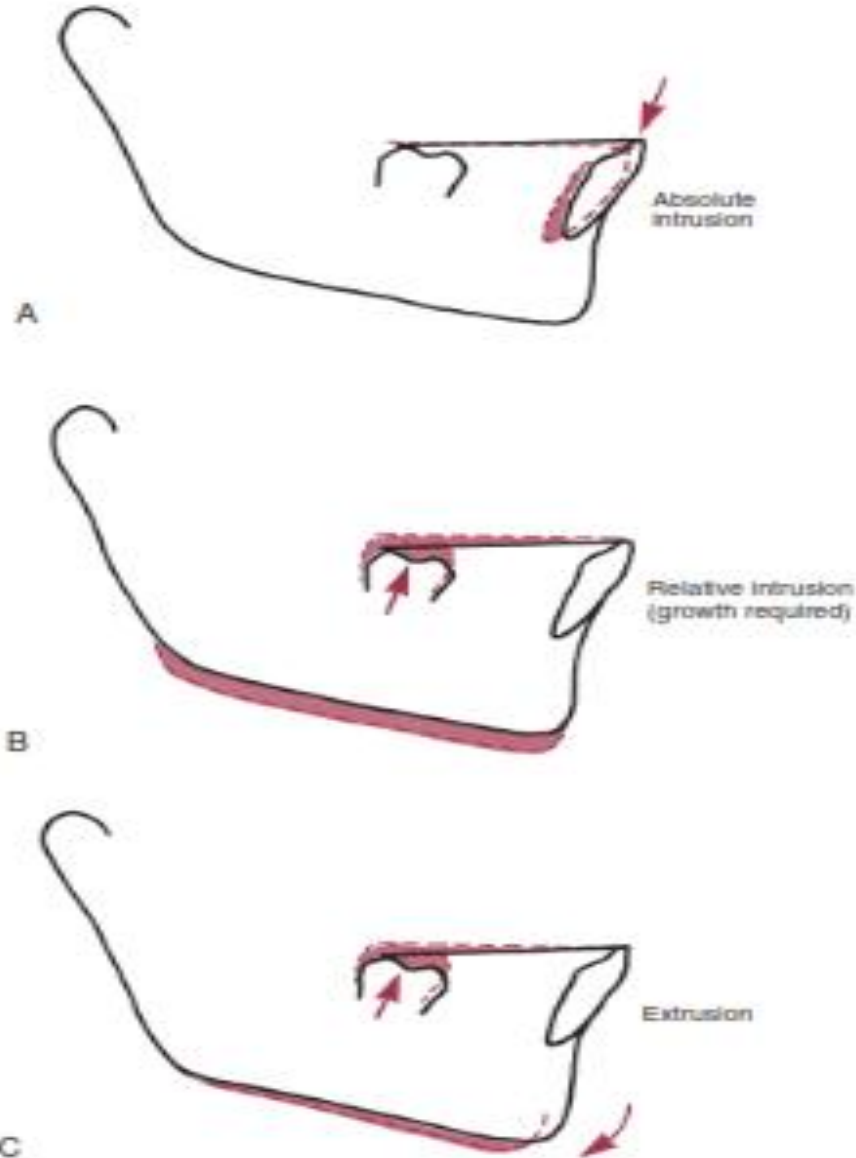
✓ **Leveling by Extrusion (Relative Intrusion)**

- Leveling by extrusion can be accomplished with continuous archwires simply by placing an exaggerated curve of Spee in the maxillary archwire and a reverse curve in the mandibular archwire.

✓ **Leveling by Intrusion**

- Leveling by intrusion with a fixed appliance requires a mechanical arrangement other than a continuous archwire attached to each tooth. The key to successful intrusion is light continuous force directed toward the tooth apex.

- There are three possible ways to level a lower arch with an excessive curve of Spee: (A) absolute intrusion; (B) relative intrusion (growth required) (C) extrusion of posterior teeth.



Cont...

- The key to successful intrusion is light continuous force directed towards the apex.
- N.B. After alignment the second arch wire is almost always 16 mil steel with an exaggerated curve of spee in the upper arch and a reverse curve of spee in the lower arch alternatively 18 mm NiTi arch wire is used.

Cont...

- ❖ Leveling requires a wire with the following specification relative to the wire used during the alignment phase of treatment:
 - Higher strength
 - Higher stiffness
 - Lower range
- ❖ Round wire is preferred for leveling because leveling does not require a CBC (counterbalancing couple).
- ❖ Couples Force system in a fixed appliance provides control in three planes of space.

The second Stage: Correction of Molar relationship and Space Closure

- ❑ At the beginning of the second stage of treatment the teeth should be well aligned & Leveled
- ❖ **The objectives of this stage of treatment are:**
 - To correct molar and buccal segment relationships
 - To provide normal occlusion in the anteroposterior plane of spaces – Class I occlusion.
 - Close extraction spaces or residual spaces in the arches and
 - Correct excessive or negative overjet.

A. Correction of Molar relationship

- ❑ The second phase in orthodontic treatment is correction of molar position.
- ❑ Therefore the goal of correction of molar position is to place the first molars in a Class I relationship.
- ❑ The process of correction of molar position is accomplished by bodily movement of the first molars into a Class I molar relationship or translation.

Cont...

- Therefore we use a Rectangular arch wire to correct molar relationship by translation; thus we have to create a CBC.
- ❑ Correction of molar position requires a wire with the following specification relative to the wire used during the leveling phase of treatment:
 - Higher strength
 - Higher stiffness
 - Lower range

B. Space closure

- This step closes any gaps that remain between the teeth after all of the teeth are aligned, leveled, and have proper molar position.
- ❑ Space closure can be accomplished in many ways.
 1. Use a wire formed into the shape of a loop.
 - The principle here is if a wire is bent out of its original position, the wire will tend to return to its original shape, provided the wire is not permanently deformed.
 - If a wire has a shape of a loop and is stretched, the wire will tend to return to the shape of a loop.
 - If a loop-shaped wire were stretched to connect two teeth at each end, then a spring back force would be created that would cause the two teeth to move together, thereby closing the space.
 2. Using Elastic chains or Elastomeric chain.

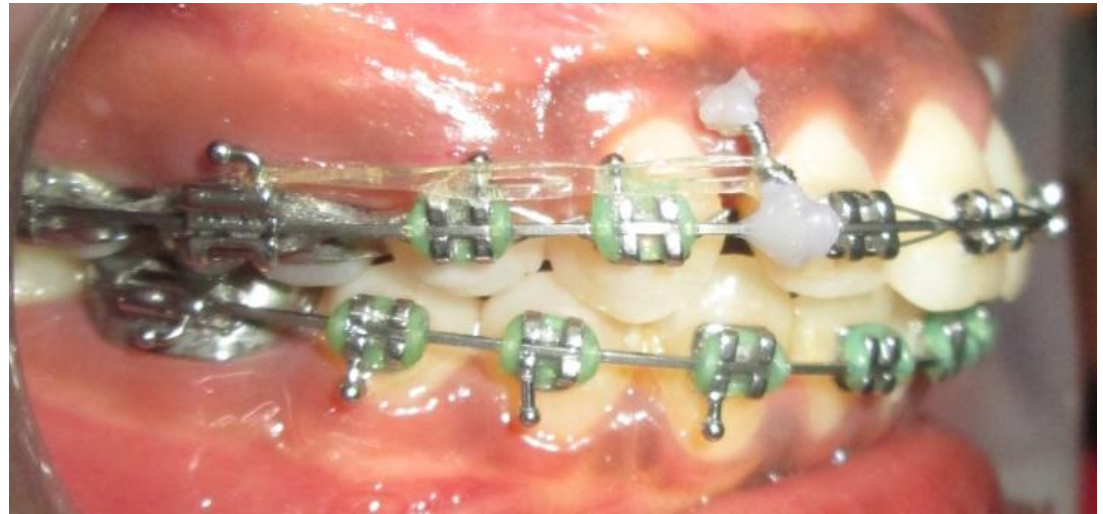
Loop Mechanics in Space Closure

❑ Closing loop



Sliding Mechanics

□ E-chain



**TABLE
16.1**

Sliding Versus Loop Mechanics

Method	Sliding teeth on an archwire	Closing loop between segments
Generation of force	Elastic or NiTi spring to single tooth or group of teeth	Activate the loop
Net force desired	100 gm per tooth	150 gm per segment
Resistance to sliding	Approximately 100 gm per tooth	None
Generation of moment	Automatic (bracket width)	Gable bend, approximately 45 degrees

NiTi, Nickel–titanium.

Cont...

- ❑ The process of space closure is accomplished by moving the teeth together to remove any gaps between the teeth.
- ❑ Space closure requires a wire with the following specification relative to correction of molar position:
 - Higher Strength
 - Higher Stiffness
 - Lower Range
- ❑ During space closure Rectangular wire is preferred because a wire with distinct edges allows us to bend the wire in the same plane of space without deviation.
- ❑ A rectangular wire also allows us to control better the position of the root during the translation of the teeth together.

The Third Stage: Finishing

- ❑ By the end of the second stage of treatment the teeth should be:
 - Well aligned and leveled
 - The teeth in the buccal segments should be in a normal Class I relationship (molar relationship should be in class I).
 - Extraction spaces should be closed
- ❑ **some adjustment of individual tooth positions needed:**
 - To get marginal ridges level;
 - To obtain precise in – out positions of teeth within the arches
 - Over come any discrepancies produced by errors in either bracket placement or appliance prescription.

Some of the procedures during finishing stage

1. Adjustment of individual tooth positions

- Some teeth may require adjustment at the finishing stage and is likely that **up – down** and **in – out** relationships of some teeth also will need minor change.
- Minor in – out and up –down adjustments typically to obtain perfect canine interdigitation and level out marginal ridge heights can be obtained simply and easily by placing mild step bends in the finishing arch wires.

Cont...

2. Correction of vertical incisor relationships

❑ Excessive Overbite

- Mildly excessive overbite at the finishing usually is treated best by slightly **intruding the maxillary incisors.**

❑ Anterior Open Bite

- Mild open bite at the end of treatment usually is treated best by **elongating the lower but not the upper incisors.**
- This is both more esthetic and more stable than elongating the upper incisors.

Cont...

3. Midline discrepancies

- The midline often can be corrected by using asymmetric Class II (or Class III) elastic force.
- As a general rule it is more effective to use Class II or Class III elastic bilaterally with heavier force on one side than to place a unilateral elastic.
- It is also possible to combine a class II or Class III elastic on one side with a diagonal elastic anteriorly to bring the midlines together.

Cont...

4. Tooth size discrepancies

- ❑ Reduction of interproximal enamel (interproximal stripping) is the usually strategy to compensate for discrepancies caused by excess tooth size.
- ❑ Tooth size discrepancy problems often are caused by small maxillary lateral incisors.
- ❑ Composite resin build up of the small lateral incisors usually is the best plan for small incisors.

5. Root Parallelizing

6. Torque



Methods for Settling the Teeth Into Ideal Occlusion

- ❑ The final step of finishing therefore is appropriately called “settling” since its purpose is to bring all the teeth into a solid occlusal relationship before the patient is placed in retention.
- ❑ Replacing full dimension rectangular wires with light round wires at the very end of treatment was the original method for settling.
- ❑ These light arches will quickly settle the teeth into final occlusion and should remain in place for only a few weeks at most.

➤ There are three ways to settle the occlusion:

- Replacing the rectangular arch wires at the very end of treatment with light round arches that provide some freedom for movement of the teeth (16 mil in the 18-slot appliance, 16 or 18 mil in the 22-slot appliance) and using light vertical elastics to bring the teeth together
- Using laced posterior vertical elastics after removing the posterior segments of the archwires
- Using atooth positioner after the bands and brackets have been removed



B

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- Using a tooth positioner

